

FACTORY OPTIMIZATION CHECKLIST

The Assembly Line · Print. Play. Optimize. Repeat.

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BEFORE YOU BUILD

- ☐ Define your end goal (rocket? colony? resource?)
- ☐ Scout the map for key resources first
- ☐ Plan a main bus or logistics spine
- ☐ Leave room: 2× what you think you'll need
- ☐ Set up starter power before factories
- ☐ Build a bootstrap setup; iterate from there

PRODUCTION RATIOS

- ☐ Check recipe inputs and outputs first
- ☐ Calculate machines needed per output
- ☐ Match belt / pipe throughput to demand
- ☐ Balance assemblers — nothing sits idle
- ☐ Use a community ratio calculator
- ☐ Round up — over capacity beats under

POWER & FUEL

- ☐ Never run at 100% power capacity
- ☐ Build power before you need it
- ☐ Diversify fuel sources when possible
- ☐ Monitor consumption vs. generation
- ☐ Automate fuel delivery to generators

COMMON MISTAKES

- ☐ Building too tight — leave expansion room
- ☐ Ignoring power until the grid browns out
- ☐ Over-optimizing before launch — just launch
- ☐ One belt carrying too many item types
- ☐ Not automating research early
- ☐ Hand-feeding furnaces past 10 minutes

THROUGHPUT & FLOW

- ☐ Don't mix different items on one belt
- ☐ Prioritize direct insertion where possible
- ☐ Use splitters and mergers intentionally
- ☐ Avoid spaghetti — bus, grid, or modules
- ☐ Buffer chests / tanks at key junctions
- ☐ Fix bottlenecks from output backward

LOGISTICS & STORAGE

- ☐ Centralize common resources
- ☐ Use requestor / provider chests smartly
- ☐ Don't over-store — buffers, not warehouses
- ☐ Label storage areas for future-you
- ☐ Automate trash / overflow disposal
- ☐ Set up a mall or hub for personal supplies

SCALING UP

- ☐ Copy-paste proven designs (blueprints)
- ☐ Modularize: one product per block
- ☐ Connect modules with logistics or trains
- ☐ Don't redesign — stamp and repeat
- ☐ Scale power first, then production
- ☐ Test at small scale before going big

DEFENSE & ENVIRONMENT

Factorio, Col, Mindustry, StarRupture.

- ☐ Wall off perimeter early
- ☐ Automate ammo / repair packs
- ☐ Turrets at chokepoints, not everywhere
- ☐ Pollution / threat = expansion cost

RESEARCH & PROGRESSION

- ☐ Automate research from the start
- ☐ Prioritize unlocks that boost throughput
- ☐ Don't hoard — spend research continuously
- ☐ Plan tech tree before committing branches

DEBUGGING STALLS

When production STOPS:

- ☐ Check the power grid first
- ☐ Walk the line from output back to input
- ☐ Look for full belts (downstream jam)
- ☐ Look for empty belts (upstream starved)
- ☐ Verify inserter / arm orientations

When production is SLOW:

- ☐ Find the slowest machine in the chain
- ☐ Add parallel machines to that step
- ☐ Upgrade belts / pipes feeding that step
- ☐ Check if you're belt-limited, not machine-limited

PER-GAME QUICK TIPS

Factorio

- ☐ Ratio everything to red belts early.

Satisfactory

- ☐ Build vertical. 1.2 changes oil — re-plan.

Shaped 2 (1.0)

- ☐ Manufacture Mode = persistent recipes.

Dyson Sphere

- ☐ Logistics towers are the breakpoint.

Foundry (Update 4)

- ☐ Towers (4–6 floors). Trains: Apr 28.

Captain of Industry

- ☐ Food before factory — always.

COMMUNITY TOOLS

Factorio

- ☐ Helmod (in-game), [kirkmcDonald.github.io](https://github.com/kirkmcDonald)

Satisfactory

- ☐ satisfactory-calculator.com, ficsit.app

Shaped 2

- ☐ Steam Workshop, shaped2.com/wiki

Dyson Sphere

- ☐ factoriolab.github.io

Captain of Industry

- ☐ captain-of-industry.com/wiki

PRO TIPS | *The factory must grow.* | *Perfect is the enemy of launched.* | *Automate anything you do twice.* | *Save before any rebuild.* | *Fun > perfect efficiency.*